

Andres Uribe Garcia

Bucaramanga, Colombia | andresfelipeuribe11@gmail.com | +57 301 656 1380

linkedin.com/in/andresfuribeg | lotuszaheer.github.io | github.com/LotusZaheer

PROFESSIONAL PROFILE

Software Developer with over 3 years of experience in mobility and logistics solutions. Specialized in geospatial data processing using Python and C++, interactive map interfaces with Angular, and optimization models using Pyomo and OR-Tools. I develop tools that transform complex spatial data into analytical and visual solutions to support operational decision-making. My work covers Angular frontend development, data flow automation, and mathematical modeling for routing and allocation problems.

EDUCATION

Universidad Industrial de Santander, Bachelor's Degree in Systems Engineering

2017 – 2022

WORK EXPERIENCE

Software Developer, OrusXpert SAS

July 2022 – Present

- Development of solutions for land transportation analysis, data visualization, and logistics optimization using Python and C++.
- Implementation of internal REST APIs with Django for complex data flow integration, including unit testing and version control with Git.
- Management and optimization of MySQL and BigQuery databases for efficient storage and querying of large data volumes.
- Development and maintenance of Angular frontend with reusable, documented components.
- Implementation of interactive maps for geospatial data visualization using Google Maps API and Folium for internal validation.

PROJECTS

Design of an IoT platform for monitoring water level and weather conditions in a river channel to provide early disaster alerts at strategic points

- Developed the platform using AWS EC2 for hosting and data processing, with MongoDB as the main database.
- Programmed ESP32, ESP8266, and Raspberry Pi devices in C++ and Python for real-time data capture and transmission.
- Integrated communication protocols (MQTT, CoAP, AMQP), ensuring reliable data flow between sensors and remote servers.
- Developed embedded software using PlatformIO, ensuring stable performance and low latency.

TECHNICAL SKILLS

Languages: C++, Python, SQL, TypeScript, Java

Technologies: Angular, Docker, ESRI, GeoPandas, Google Maps API, Numpy, OR-Tools, Pandas, Polars

LANGUAGES

Spanish: Native

English: B2